# Linux for DevOps

Lesson 5: Process & Service Management

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## Process & Service Management - Study Notes

### Key Concepts

1. **Systemd Unit Files** – Declarative configuration files that describe how services are started, stopped, and monitored.

2. **Service Control Commands** – `systemctl` is the primary CLI for enabling, disabling, starting, stopping, and querying service status.

3. **Process Monitoring Tools** – `ps`, `top`, and `htop` provide snapshots or live views of running processes, CPU/memory usage, and other metrics.

4. **Process Hierarchy & PIDs** – Understanding parent/child relationships, session leaders, and process groups is essential for troubleshooting.

5. **Logging & Journaling** – `journalctl` works with systemd to collect logs, making it easier to diagnose service failures.

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### Summary

Systemd is the modern init system used by most Linux distributions. It replaces older SysV init scripts with declarative unit files that define how a service should run, its dependencies, and its runtime environment. The `systemctl` command is the main interface for interacting with these units: you can start, stop, restart, enable (run at boot), disable, and check the status of services.

Monitoring processes is a complementary skill. `ps` gives a static snapshot of processes, `top` offers a continuously updating view of CPU/memory usage, and `htop` enhances `top` with color, tree view, and easier interaction. Together, these tools let you verify that services are running, identify resource hogs, and debug issues.

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### Important Points

- **Unit Types**

- `service` – a long  running daemon.
- `socket` – activates a service on demand.
- `target` – groups units for ordering.

- **Common `systemctl` Options**

- `start|stop|restart|reload|enable|disable|status|is-enabled|is-active`
- `systemctl daemon-reload` after editing unit files.

- **Process Identification**

- `ps -ef` – full-format list.
- `ps -o pid,ppid,user,cmd,%cpu,%mem` – custom columns.
- `pgrep -u username` – filter by user.

- **Top & Htop Usage**
  - `top` – press `P` to sort by CPU, `M` for memory.
  - `htop` – use arrow keys, `F6` to sort, `F9` to kill.
- **Logging**
  - `journalctl -u servicename` shows logs for a specific unit.
  - `journalctl -b` shows logs since the last boot.
- **Permissions**
  - Most service control requires `sudo`.
  - `systemctl --user` manages user level services.

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## Examples

### 1. Enabling and Starting a Service

```bash
# Enable nginx so it starts on boot
sudo systemctl enable nginx.service
# Start it immediately
sudo systemctl start nginx.service
# Verify status
sudo systemctl status nginx.service
```

*Explanation:* `enable` writes symlinks to the boot target, `start` launches the daemon, and `status` shows active state, recent logs, and PID.

### 2. Monitoring a Process Tree with `htop`

```bash
# Launch htop
htop
# Press 'F5' to view the process tree
# Use arrow keys to navigate, 'F9' to send signals (e.g., kill)
```

*Explanation:* The tree view shows parent/child relationships, making it easy to spot runaway processes or orphaned children that may need cleanup.

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## Quick Review

1. What command would you use to reload systemd after editing a unit file?
2. How can you view the last 50 lines of the system journal for the `sshd` service?
3. Which `ps` option lists processes in a tree format?
4. In `htop`, which function key allows you to sort by memory usage?
5. Explain the difference between `systemctl enable` and `systemctl start`.

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## Further Reading

- **Systemd Unit File Syntax** – Dive deeper into directives like `ExecStart`, `Restart`, `EnvironmentFile`.
- **Socket Activation** – Learn how `systemd` can start services on demand.
- **Process Accounting** – Tools like `acct`, `psacct`, and `systemtap`.
- **Advanced `top`/`htop` Customization** – Adding custom columns, scripting.
- **Systemd Logging** – Using `journalctl` filters, persistent storage, and log rotation.